VitaRoute

Intelligent Organ Matching and Critical Transit System

Core technique: Constraint Satisfaction Problem with AC-3 and Backtracking

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Abstract

VitaRoute is a simulated organ matching and critical transit system that models organ allocation as a Constraint Satisfaction Problem (CSP). Patients are the variables and organs are their possible values.



Eight constraints are enforced by AC-3 before search: organ type, blood group, HLA, viability, hospital capability, transit time, consent and availability.



Backtracking Search finds a valid patient–organ assignment in urgency order and never forces a match.



A transit module confirms that delivery fits inside the remaining organ viability.



The prototype uses fictional data and no machine learning.

Problem Statement

Organs stay viable for only a few hours, so every minute of delay reduces the chance of a transplant.

Many conditions must hold together: organ type, blood group, HLA match, hospital capability, consent and travel time.

Checking these by hand across hospitals is slow, error-prone and hard to explain.

Problem: how can a system find a valid organ for a patient quickly, explain why, and confirm the organ can reach the hospital in time?



Literature Survey

Approach	Idea	Limitation
National allocation registries (OPTN/UNOS, Eurotransplant, NOTTO)	Waiting lists with priority rules	Policy-driven; transit feasibility is handled separately
Optimisation-based exchange (graph or integer programming)	Optimal paired-donation matching	Heavy computation; hard to explain step by step
Machine-learning outcome models	Predict transplant success from clinical data	Needs large real datasets; works like a black box
CSP-based scheduling and allocation	Constraints and search for assignment problems	Not usually applied together with transit viability

Research Gap

01

Separate Processes

Matching and transit feasibility are often handled as separate steps, making it difficult to verify whether a selected organ can actually reach the hospital in time.

02

Limited Explainability

Optimisation and machine-learning approaches often do not clearly explain why an organ was accepted or rejected.

03

Lack of Step-by-Step Reasoning

Existing approaches provide limited visibility into domain reduction, constraint checking, and search decisions.

Research Need

An explainable, constraint-based method that links organ matching with transit feasibility and **clearly** shows the reasoning behind every **decision.**



Research Objectives



1. Model organ matching as a CSP with variables, domains and constraints.
2. Prune inconsistent organs with AC-3 before search.
3. Find valid assignments with Backtracking Search in urgency order.
4. Check organ viability and transit feasibility before confirming a match.
5. Show the reasoning visually and report clear reasons when no match exists.

VitaRoute System Architecture

Intelligent Organ Matching and Critical Transit System

1. USERS AND PRESENTATION

HTML, CSS, JavaScript

Admin
Manage system and view reports

Patient Hospital
Register patients and view matches

Donor Hospital
Register organs and track transit

User Interface (Web Application)

- Login and Role Selection
- Dashboard
- Register Patient
- Register Organ
- Run Intelligent Matching
- Match Result
- Transit Tracking
- CSP Intelligence Center (AC-3 steps and Backtracking replay)

User Requests

2. SERVICES

Application Logic (Node.js / Express.js)

Authentication and Role Access

- User login
- Role-based access control

Patient and Organ Registration

- Register / manage patients
- Register / manage organs
- Validate inputs

Matching Service

- Build CSP with registered data
- Call AC-3 and Backtracking
- Return match result

Transit Service

- Calculate distance and time
- 7-stage transit status
- Check feasibility with viability

Status Updates and Notifications

- Update match and transit status
- Notify hospitals

Processed Data

3. CSP ENGINE (Core AI)

CSP → AC-3 → Backtracking → Organ Matching → Critical Transit Decision

1. **Build CSP**
   Variables: Patient, Organ, Hospital, Transit
   Domain: all registered organs
2. **Constraints**
   - Organ type
   - Blood group
   - HLA compatibility
   - Organ viability
   - Hospital capability
   - Transit feasibility (time)
   - Consent
   - Availability
   - All-different (one organ per patient)
3. **AC-3 (Arc Consistency)**
   - Initialize arc queue
   - Revise domains using constraints
   - Domain reduction (remove inconsistent organs)
4. **Backtracking Search**
   - Select next patient (urgency order)
   - Try an organ
   - On conflict, backtrack and try next organ
   - Assign if all constraints satisfied
5. **Transit Feasibility Check**
   - Calculate transit time
   - Transit time < remianing organ viability
6. **Match Result / No Feasible Match**
   - Valid patient–organ assignment
   - If no match, provide clear reasons

Read / Write Data

4. DATA

Database (SQLite / In-memory)

Hospital

- Hospital ID
- Name, Location
- Organ capabilities

Patient

- Patient ID
- Name, Blood group
- Required organ
- Urgency level
- Hospital

Organ

- Organ ID
- Type, Blood group
- HLA details
- Donor hospital
- Viability time
- Availability status

Match

- Match ID
- Patient ID
- Organ ID
- Status, Timestamp
- Reasons (if no match)

Transit

- Transit ID
- Match ID
- Source and destination
- Estimated time
- 7-stage status
- Current status, Timestamp

Match Result, Reasons and Transit Updates (Displayed to Users)

Legend: Data Flow · Database Read/Write · Result / Update Flow · Core AI Component

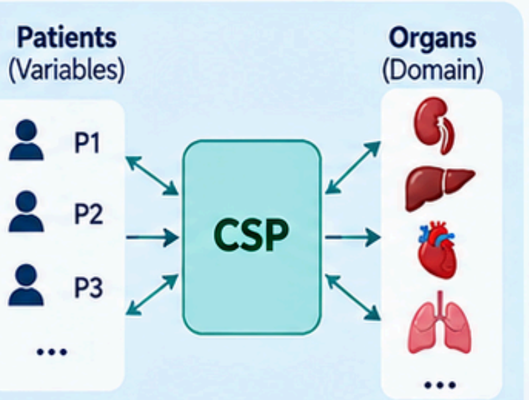
Simulation prototype – not a clinical decision system
Uses fictional data for demonstration purposes only.

Proposed Work – Algorithm

CSP → AC-3 → Backtracking → Organ Matching → Critical Transit Decision

1 Build the CSP

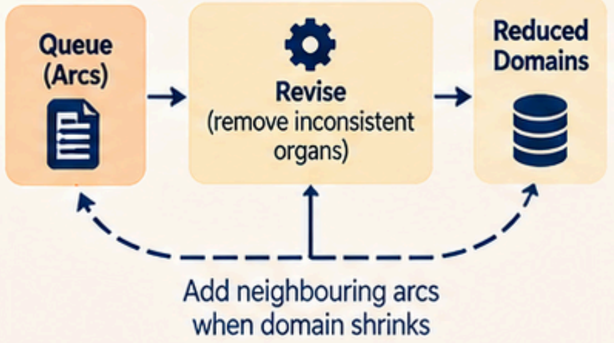
Each patient is a variable.
Every organ starts in its domain.



Variables: Patient, Organ, Hospital, Transit
Domain: All registered organs for each patient

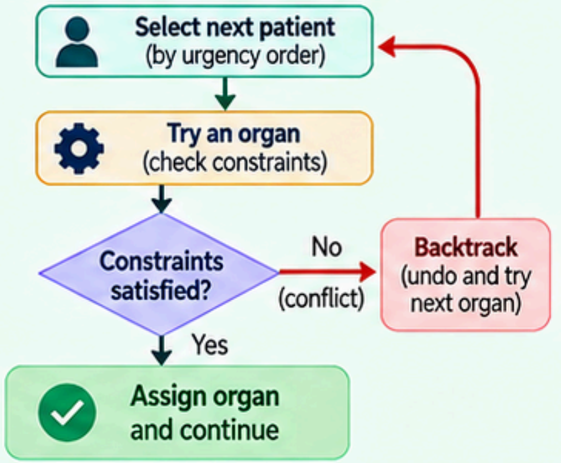
2 AC-3 (Arc Consistency)

- Put all arcs (patient, constraint) in a queue.
- Revise: remove organs that violate the constraint.
- When a domain shrinks, add the neighbouring arcs back to the queue.
- Stop when the queue is empty; return the reduced domains.



3 Backtracking Search

- Pick the next patient in urgency order:
- Critical, High, Medium, Low.
- Try an organ, check the constraints, assign it, then recurse.
- On a conflict, undo the assignment and try the next organ.
- If no organ is left, leave the patient unmatched (never force a match).



4 Transit Check

Transit time must be less than the remaining organ viability.



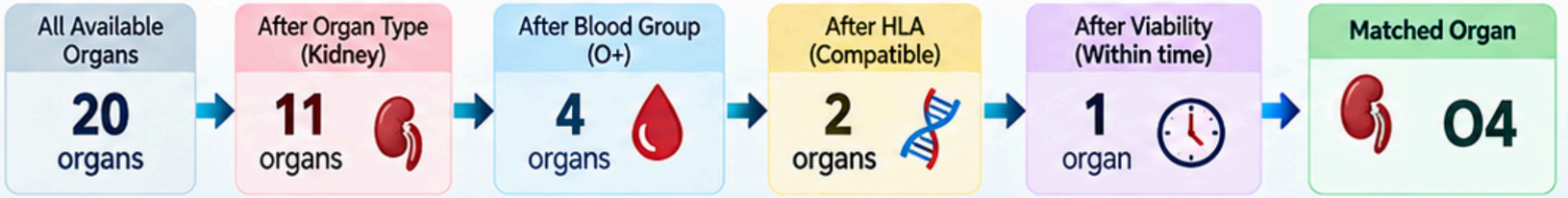


Match Result
(Valid patient–organ assignment)

No Feasible Match
(Provide clear reasons)

Worked Example – Arun Kumar (VT-P1001)

Patient Details: Arun Kumar | ID: VT-P1001 | Blood Group: O+ | Organ: Kidney | Urgency: **Critical**



Why Both?

AC-3	Reduces the search space by removing inconsistent organs.
Backtracking	Finds the final valid assignment from the reduced search space.

Implementations And Limitations

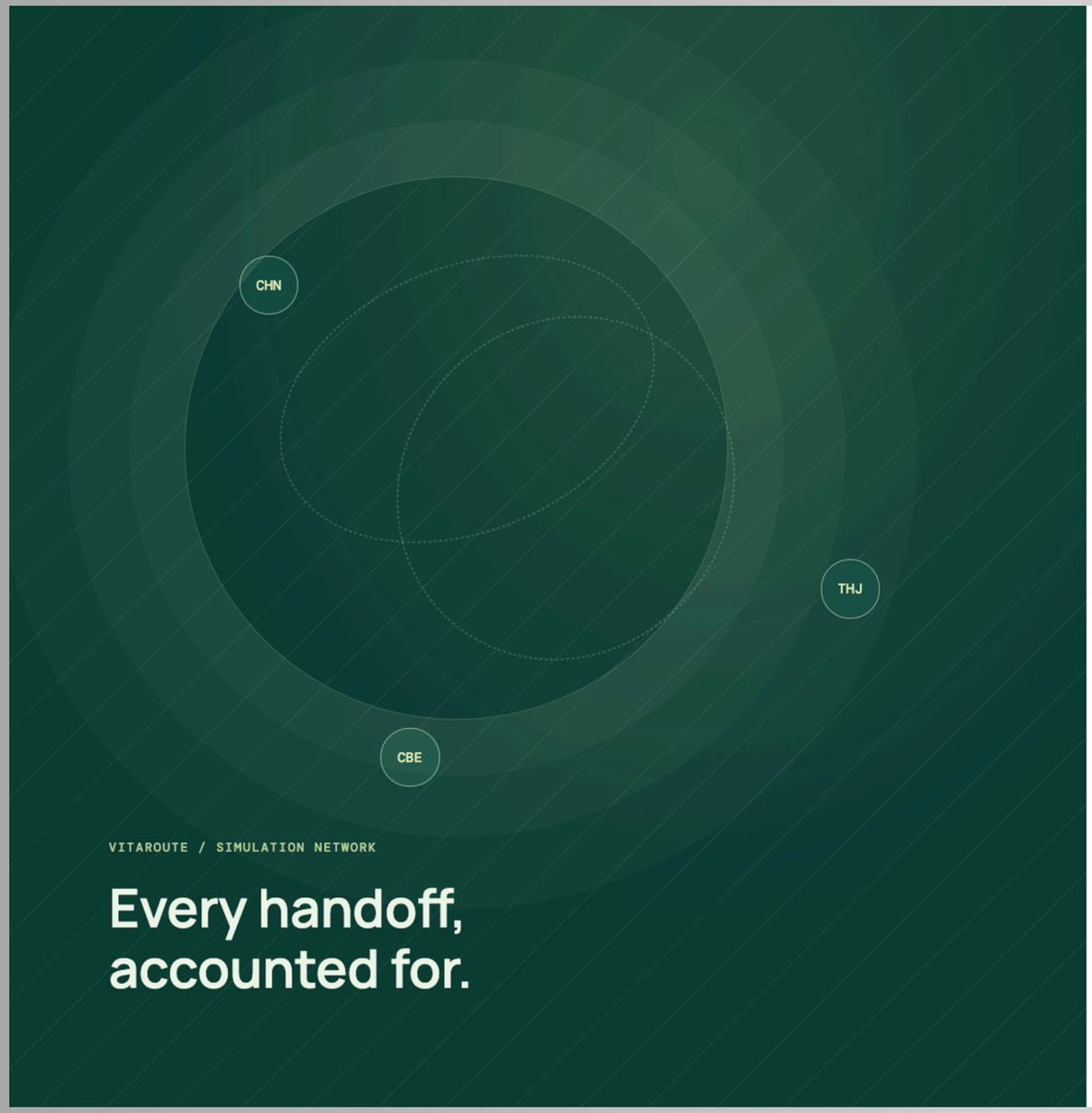
Implemented

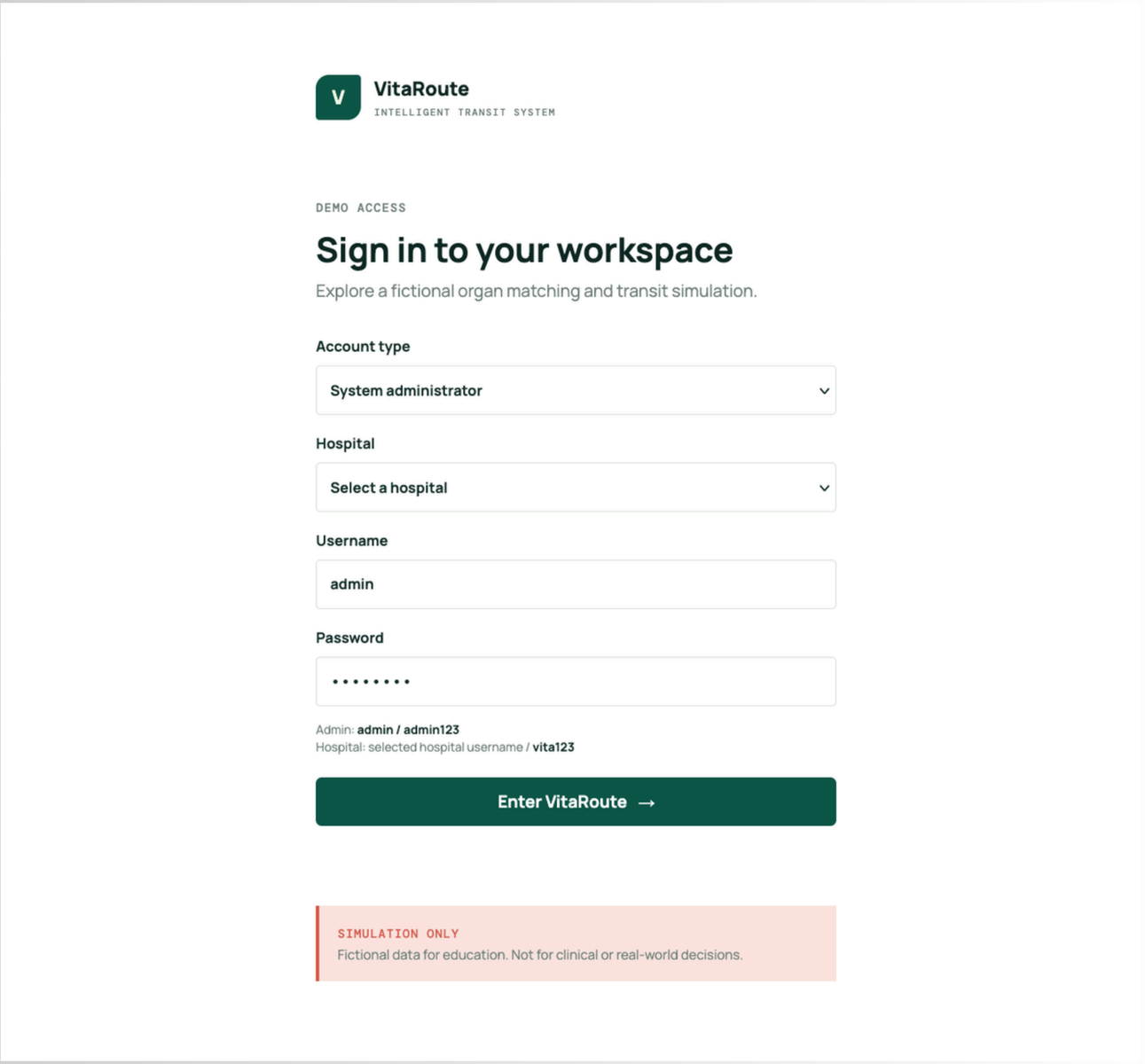
- Single-page HTML/JavaScript application with Admin, Patient Hospital and Donor Hospital dashboards.
- Sample data: 10 hospitals across Tamil Nadu, 20 patients, 20 organs (all fictional).
- AC-3, Backtracking and the matching engine, kept separate from the user interface.
- Automatic transit record with a 7-stage timeline.
- CSP Intelligence Center with an AC-3 trace and a Backtracking replay.

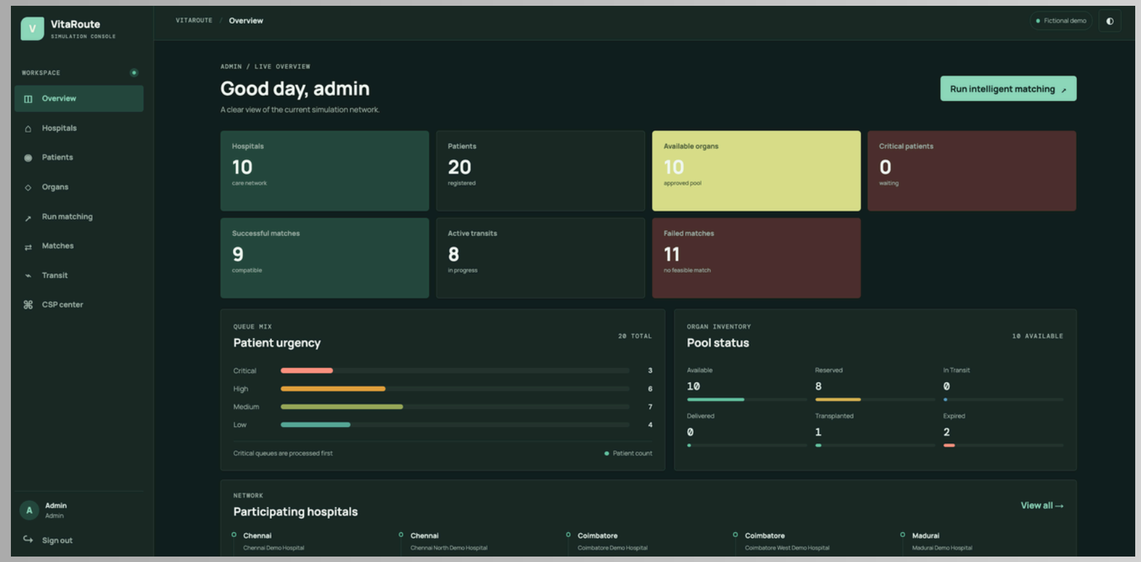
Limitations

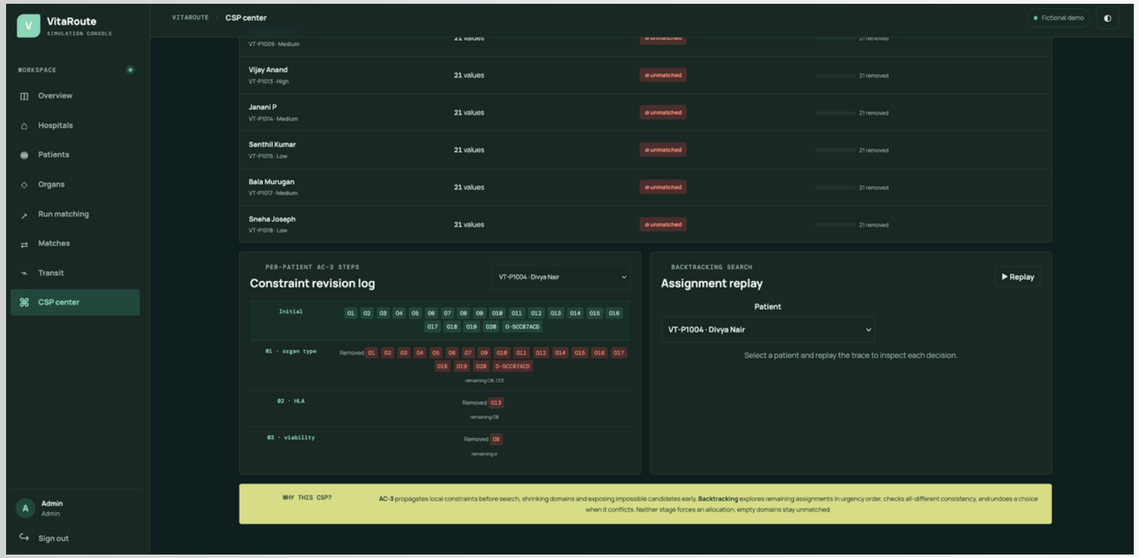
- Simulation only; not a clinical decision system.
- Blood group and HLA rules are simplified.
- Transit time comes from a distance formula, not real roads.
- Data is stored in memory and resets on reload; a Node.js + SQLite backend is future work.
- Patients are matched in urgency order, so the result is not globally optimal.

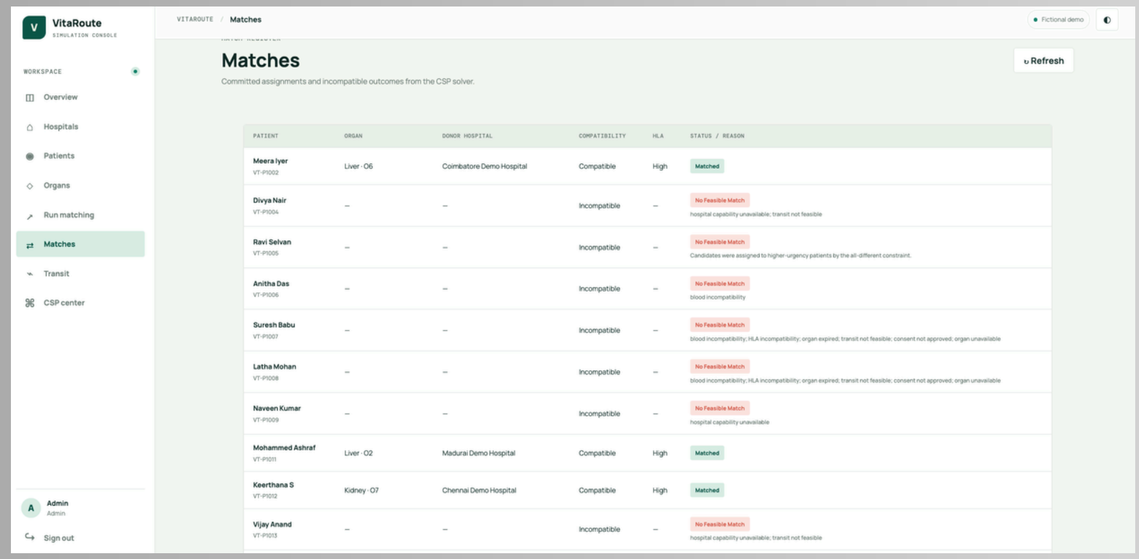
Outputs And Result

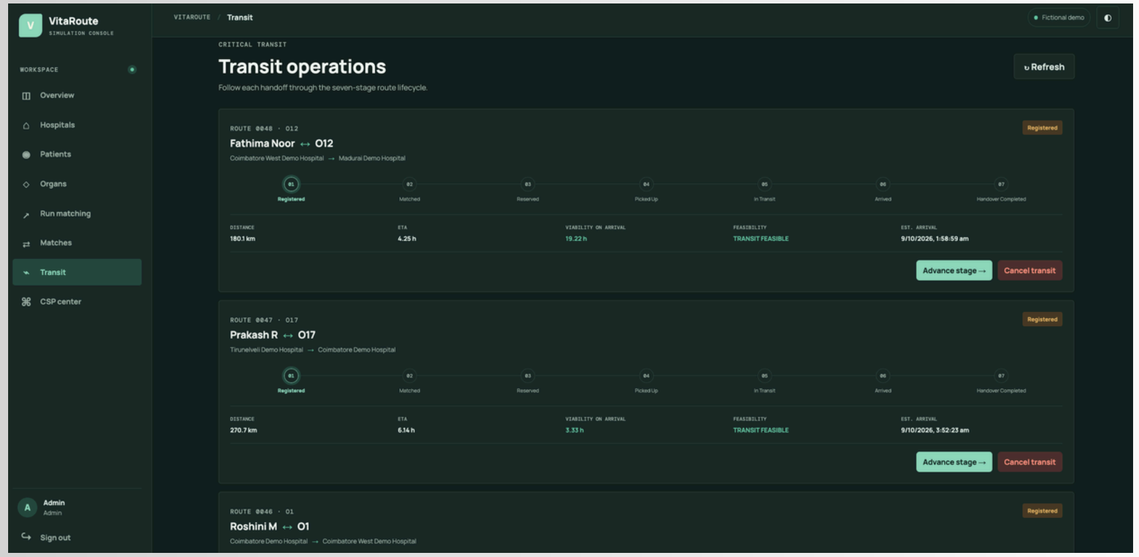





VitaRoute
SIMULATION CONSOLE
VITAROUTE / Overview
Fictional demo
WORKSPACE
Overview
Hospitals
Patients
Organs
Run matching
Matches
Transit
CSP center
ADMIN / LIVE OVERVIEW
Good day, admin
A clear view of the current simulation network.
Run intelligent matching
Hospitals
10
care network
Patients
20
registered
Available organs
10
approved pool
Critical patients
0
waiting
Successful matches
9
compatible
Active transits
8
in progress
Failed matches
11
no feasible match
QUEUE MIX
Patient urgency
20 TOTAL
Critical 3
High 6
Medium 7
Low 4
Critical queues are processed first
Patient count
ORGAN INVENTORY
Pool status
10 AVAILABLE
Available 10
Reserved 8
In Transit 0
Delivered 0
Transplanted 1
Expired 2
NETWORK
Participating hospitals
View all →
Chennai
Chennai Demo Hospital
Chennai
Chennai North Demo Hospital
Coimbatore
Coimbatore Demo Hospital
Coimbatore
Coimbatore West Demo Hospital
Madurai
Madurai Demo Hospital
Admin
Admin
Sign out


VitaRoute
SIMULATION CONSOLE
VITAROUTE CSP center
Fictional demo
WORKSPACE
Overview
Hospitals
Patients
Organs
Run matching
Matches
Transit
CSP center
Vijay Anand
VT-P1003 · High
21 values
unmatched
21 removed
Janani P
VT-P1014 · Medium
21 values
unmatched
21 removed
Senthil Kumar
VT-P1015 · Low
21 values
unmatched
21 removed
Bala Murugan
VT-P1017 · Medium
21 values
unmatched
21 removed
Sneha Joseph
VT-P1018 · Low
21 values
unmatched
21 removed
PER-PATIENT AC-3 STEPS
Constraint revision log
VT-P1004 · Divya Nair
Initial
01 · organ type
02 · HLA
03 · viability
BACKTRACKING SEARCH
Assignment replay
Replay
Patient
VT-P1004 · Divya Nair
Select a patient and replay the trace to inspect each decision.
WHY THIS CSP?
AC-3 propagates local constraints before search, shrinking domains and exposing impossible candidates early. Backtracking explores remaining assignments in urgency order, checks all-different consistency, and undoes a choice when it conflicts. Neither stage forces an allocation; empty domains stay unmatched.
Admin
Admin
Sign out


VitaRoute
SIMULATION CONSOLE
VITAROUTE / Matches
Fictional demo
WORKSPACE
Overview
Hospitals
Patients
Organs
Run matching
Matches
Transit
CSP center
Matches
Committed assignments and incompatible outcomes from the CSP solver.
Refresh
PATIENT
ORGAN
DONOR HOSPITAL
COMPATIBILITY
HLA
STATUS / REASON
Meera Iyer
VT-P1002
Liver · O6
Coimbatore Demo Hospital
Compatible
High
Matched
Divya Nair
VT-P1004
Incompatible
No Feasible Match
hospital capability unavailable; transit not feasible
Ravi Selvan
VT-P1005
Incompatible
No Feasible Match
Candidates were assigned to higher-urgency patients by the all-different constraint.
Anitha Das
VT-P1006
Incompatible
No Feasible Match
blood incompatibility
Suresh Babu
VT-P1007
Incompatible
No Feasible Match
blood incompatibility; HLA incompatibility; organ expired; transit not feasible; consent not approved; organ unavailable
Latha Mohan
VT-P1008
Incompatible
No Feasible Match
blood incompatibility; HLA incompatibility; organ expired; transit not feasible; consent not approved; organ unavailable
Naveen Kumar
VT-P1009
Incompatible
No Feasible Match
hospital capability unavailable
Mohammed Ashraf
VT-P1011
Liver · O2
Madurai Demo Hospital
Compatible
High
Matched
Keerthana S
VT-P1012
Kidney · O7
Chennai Demo Hospital
Compatible
High
Matched
Vijay Anand
VT-P1013
Incompatible
No Feasible Match
hospital capability unavailable; transit not feasible
Admin
Admin
Sign out


VitaRoute
SIMULATION CONSOLE
VITAROUTE / Transit
Fictional demo
WORKSPACE
Overview
Hospitals
Patients
Organs
Run matching
Matches
Transit
CSP center
CRITICAL TRANSIT
Transit operations
Follow each handoff through the seven-stage route lifecycle.
Refresh
ROUTE 0048 · O12
Fathima Noor ↔ O12
Coimbatore West Demo Hospital → Madurai Demo Hospital
Registered
Matched
Reserved
Picked Up
In Transit
Arrived
Handover Completed
DISTANCE
180.1 km
ETA
4.25 h
VIABILITY ON ARRIVAL
19.22 h
FEASIBILITY
TRANSIT FEASIBLE
EST. ARRIVAL
9/10/2026, 1:58:59 am
Advance stage →
Cancel transit
ROUTE 0047 · O17
Prakash R ↔ O17
Tirunelveli Demo Hospital → Coimbatore Demo Hospital
DISTANCE
270.7 km
ETA
6.14 h
VIABILITY ON ARRIVAL
3.33 h
FEASIBILITY
TRANSIT FEASIBLE
EST. ARRIVAL
9/10/2026, 3:52:23 am
Advance stage →
Cancel transit
ROUTE 0046 · O1
Roshini M ↔ O1
Coimbatore Demo Hospital → Coimbatore West Demo Hospital
Admin
Admin
Sign out

VitaRoute

Intelligent Organ Matching and Critical Transit System

Thank You

FOR YOUR TIME AND ATTENTION



- Donate Organs
- Save Lives
- Stronger Communities
- A Healthier Tomorrow

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Simulation prototype – not a clinical decision system